

Confidence-Gated Explainable AI for Reliable Churn Prediction and Automated Retention Decisions

FINAL YEAR

PROJECT PHASE - 1 REPORT

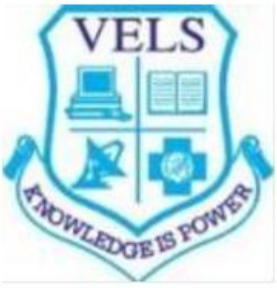
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BONAFIDE CERTIFICATE

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ABSTRACT

In B2B SaaS, the loss of even a few high-value clients can significantly impact revenue, customer lifetime value, and long-term growth, making churn a critical challenge. Existing systems for churn prediction, typically based on traditional machine learning methods such as logistic regression and tree-based models, provide low-level risk scores that do not have a clear structure for easy interpretation and have no mechanism for timely action. In this project, a framework is developed for explainable AI-based churn prediction and decision support which combines machine learning with explainability and confidence-based automation to improve customer retention. It collects and merges several data types such as account data, product usage, billing and subscriptions, support interactions, ticket activity, and CRM engagement. A data preprocessing pipeline guarantees the data quality by providing missing value imputation, outlier detection, and consistency checking. Behavioral signals like usage drops, payment delays, unclosed support tickets, and decreased engagement over time are captured. The XGBoost predictive model is used to estimate churn probability at the account level for a set time horizon. To overcome the interpretability gap, SHAP is used, which gives both global and local explanations by figuring out the most influential features of each prediction. Further analysis involves feature interaction effects and stabilizing explanations in similar accounts. A confidence-gated decision layer performs prediction classification based on the confidence level of the model and quality level of the data, as either automated intervention, human review, or abstention, giving reliability and avoiding inappropriate automated operations. Central to this framework is an AI reasoning agent that goes beyond flagging at-risk accounts to construct the underlying rationale: reasoning over SHAP attributions, account history, and behavioral context to explain why a customer is likely to churn and what specifically the organization can do to retain them, rather than leaving that interpretation to the end user.

Keywords -- XGBoost, SHAP Analysis, AI Reasoning Agent, Human-in-the-Loop, Gradient Boosting.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
B2B	Business-to-Business
API	Application Programming Interface
CSV	Comma-Separated Values
CRM	Customer Relationship Management
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
XGBoost	Extreme Gradient Boosting
SHAP	SHapley Additive exPlanations
SaaS	Software-as-a-Service
SMTP	Simple Mail Transfer Protocol
SMOTE	Synthetic Minority Over-sampling Technique

CHAPTER 1
INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Introduction

Software-as-a-Service has become an important model for delivering enterprise software through subscription-based services. In a B2B SaaS environment, customer retention is particularly important because losing a business customer can result in significant revenue loss, reduced customer lifetime value and increased acquisition costs.

Customer churn prediction aims to identify customers who are likely to discontinue their subscription within a predefined time period. Machine learning models can learn patterns from historical customer information and generate a churn probability for each account. However, a prediction alone does not provide sufficient information for a customer-success team to understand the reason behind the risk or determine the appropriate action.

For example, an account may receive a high churn probability because of declining product usage, delayed payments, unresolved support tickets, subscription changes or reduced engagement. Conventional models may identify the risk without clearly explaining the underlying factors. The project therefore combines predictive modelling with Explainable AI and reasoning-based decision support.

The proposed system uses customer-level data and performs systematic data profiling before prediction. The updated pipeline detects the target variable, identifies identifiers and useful features, checks for leakage and redundancy, generates data-quality warnings and performs feature analysis and selection.

The selected machine learning models are evaluated using cross-validation, ROC-AUC and F1-score along with other classification metrics. XGBoost serves as the primary predictive model. SHAP then explains the contribution of individual features to each prediction.

The system extends this prediction layer using AI-based interpretation and behavioral analysis. An AI reasoning layer interprets the model evidence together with customer behaviour and produces

a human-readable churn explanation. The evidence validation stage then checks whether the generated interpretation is supported by the available customer data.

Finally, a decision mechanism determines whether the case is suitable for automated intervention, requires human review or should be abstained from because the available evidence is insufficient.

Problem Statement

Existing churn prediction systems have several limitations:

- They primarily focus on predicting churn probability.
- Individual predictions may be difficult to interpret.
- Important behavioral changes may not be clearly presented.
- AI-generated explanations may contain unsupported claims if not validated.
- Prediction systems generally do not directly generate retention actions.
- Not every prediction is reliable enough for automated intervention.

Therefore, a reliable system is required that connects prediction → explanation → behavioral analysis → evidence validation → decision support.

Objectives of the Project

The major objectives are:

- To develop a machine learning-based customer churn prediction system.
- To perform systematic data profiling and quality assessment.
- To detect possible data leakage and redundant variables.
- To compare Logistic Regression, Random Forest and XGBoost.
- To use XGBoost for structured customer churn prediction.
- To generate churn probability and risk levels.

- To provide global and individual explanations using SHAP.
- To analyse customer behavioural patterns.
- To use an AI reasoning layer for human-readable interpretation.
- To validate generated evidence before decision support.
- To route cases towards automation, human review or abstention based on reliability.
- To generate actionable retention recommendations.

Scope of the Project

1.2 SCOPE AND OBJECTIVE

Purpose of the Study

The purpose of this project is to transform conventional churn prediction from a simple classification problem into a more reliable and interpretable decision-support system.

The framework does not stop after identifying a high-risk customer. Instead, it attempts to answer:

- Who is likely to churn?
- What is the probability of churn?
- What factors contributed to the prediction?
- What behavioural changes are visible?
- What is the likely reason for churn?
- Is the evidence reliable?
- Should the system act automatically?
- Should a human re
- view the case?

- What retention action can be recommended?

This approach is aligned with the research gap identified in the source paper, where prediction, explanation, confidence assessment, human oversight and actionable retention reasoning are generally handled separately.

Functional Scope

• Gathering and processing of customer information from CSV datasets, including account, usage, billing, subscription, support and engagement information.
• Automated data profiling, target detection and column-role identification.
• Detection of identifier columns, useful features, data leakage, redundant variables and data-quality issues.
• Churn prediction using Logistic Regression, Random Forest and XGBoost models.
• Generation of individual churn prediction, churn probability and risk level.
• SHAP-based global and individual customer-level explanations.
• Behavioral analysis to identify categorical patterns, numerical patterns, churn rates and effect sizes.
• AI-based interpretation using an Explainable AI Agent and Qwen.
• Evidence validation to verify numbers, features, categories and unsupported claims before producing the final report.
• Generation of a final holistic churn report with prediction, explanation, behaviour, AI interpretation, validation and decision support.

Technical Scope

• Data: Customer CSV dataset containing account information, product usage, billing/subscription records, support interactions, ticket activity and customer engagement information.
• Machine Learning: Logistic Regression, Random Forest and XGBoost for comparative churn prediction.
•Explainable AI: SHAP for global and individual feature-level explanations.
• AI Interpretation: Qwen for converting structured analytical evidence into human-readable

interpretations.

•Data Processing: Python, Pandas, NumPy and Scikit-learn.

•Model Development: XGBoost and Scikit-learn-based machine learning pipeline.

• Validation: Cross-validation, ROC-AUC, F1-score, precision, recall and evidence validation.

• Decision Support: Confidence/reliability-based decision mechanism for automated action, human review or abstention.

Application Scope

The system is applicable in:

• B2B SaaS Customer Retention: Identification of business customers who are at risk of discontinuing their subscriptions.

• Customer Success Management: Helps customer-success teams understand why an account is at risk and prioritize intervention.

• Subscription Businesses: Supports prediction and prevention of customer subscription cancellations.

• Business Analytics: Provides population-level behavioral insights for understanding common churn patterns.

System Overview

Data Preparation and Intelligence Layer:

• The system loads the customer dataset from a CSV file.

• Data profiling is performed to understand the structure, data types, missing values and quality of the dataset.

• The target variable is detected automatically.

• Columns are analysed to identify identifiers, useful features, potential leakage and redundant variables.

• Data-quality warnings are generated before model training.

Feature Engineering and Prediction Layer:

• Feature analysis and selection are performed to identify relevant customer attributes.
• The dataset is divided into training and testing sets.
• Missing values are handled, numerical features are scaled where required and categorical features are encoded.
• Logistic Regression, Random Forest and XGBoost models are trained and compared.
• The best-performing model is selected based on the evaluation results.
• The selected model generates churn prediction, churn probability and risk level.

Explainability Layer:

• SHAP is used to explain the model's predictions.
• Global explanations identify the most influential churn features across the customer population.
• Local explanations identify the strongest factors influencing an individual customer's prediction.
• Individual customer evidence is extracted from the SHAP results and passed to the AI reasoning layer.

AI and Behaviour Intelligence Layer:

• The Explainable AI Agent processes the model evidence and customer information.
• Qwen interprets the structured evidence and generates a human-readable explanation.
• Behaviour analysis identifies overall churn rates, categorical patterns, numerical patterns and effect sizes.
• The Behaviour Intelligence Agent combines customer-specific evidence with population-level behavioral information.
• Reliability warnings are considered during the interpretation process.

Evidence Validation and Decision Layer:

• The Evidence Validator checks numbers, features and categories used in the generated explanation.
• The output is classified as PASS or FAIL.
• Validated information is used to generate the final churn report.

• The final system provides decision support based on the reliability of the prediction and supporting evidence.

Advantages of the System

• Reliable Prediction: Compares multiple machine learning models before selecting the best-performing model.
• Explainability: SHAP provides understandable reasons behind individual churn predictions.
• Behavioral Intelligence: Analyses customer behaviour rather than relying only on a single prediction score.
• Actionable Insights: Converts churn predictions into useful decision-support information.
• Human Oversight: Uncertain or unreliable cases can be identified for human review instead of blindly automating decisions.
• Scalability: The modular architecture can be applied to different customer datasets and SaaS environments.
• Data Quality Awareness: Leakage, redundancy and other data-quality problems are identified before model development.

Summary

• Individual customer churn probability and risk-level prediction.
• Global and local SHAP-based explanations.
• Behavioral analysis of customer churn patterns.
• Qwen-based human-readable AI interpretation.
• An evidence-validation mechanism producing PASS/FAIL validation.
• A final holistic churn report containing prediction, probability, risk level, SHAP explanation, behavioral information, AI interpretation, evidence validation and decision support.
• A reliable foundation for automated retention decisions and human-in-the-loop review.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE OVERVIEW

Recent research in customer churn prediction has progressed from traditional statistical approaches towards machine learning, ensemble learning, deep learning, temporal modelling and Explainable AI. However, the main focus of many studies remains predictive performance.

Research has demonstrated the effectiveness of models such as XGBoost, Random Forest, Transformer-based architectures and hybrid deep learning methods. Other studies have focused on class imbalance, temporal behaviour, survival analysis, feature selection and explainability.

Explainable AI approaches, particularly SHAP, provide insight into the features influencing individual predictions. Temporal approaches additionally examine how customer behaviour changes over time. CRM-oriented research focuses on converting churn predictions into customer-retention strategies.

However, these capabilities are often developed independently. The source research identifies a clear gap in integrating prediction, explanation, confidence assessment, human oversight and actionable retention reasoning within one framework.

2.2 LITERATURE SURVEY

Machine Learning and Ensemble-Based Churn Prediction

[1] Roy, Singh and Shaw proposed a customer churn prediction framework for structured tabular data using an FT-Transformer and stacking ensemble architecture. The approach combines FT-Transformer with XGBoost and a Logistic Regression meta-learner to capture attention-based feature interactions and nonlinear relationships in customer data. Class imbalance is addressed through class-weighted learning, while probability calibration is used to improve the reliability of predicted churn probabilities. The study demonstrates that hybrid modelling can improve predictive performance and probability calibration compared with individual baseline models. However, the framework mainly focuses on prediction and probability reliability and does not

provide a complete mechanism for converting reliable predictions into automated, context-specific retention actions.

[2] Imani, Joudaki, Beikmohammadi and Arabnia presented a systematic review of recent machine learning and deep learning approaches for customer churn prediction. The review analyses datasets, algorithms, evaluation methods and major challenges in churn prediction research. It highlights the increasing adoption of ensemble and deep learning techniques while identifying persistent issues such as class imbalance, limited interpretability, inconsistent evaluation practices and difficulties in deploying churn models in real-world environments. The study emphasizes the need for transparent, reliable and practically actionable churn prediction systems. However, the work is primarily a systematic review and does not itself provide an integrated prediction, explanation and retention-decision framework.

[3] Hermawan, Saputra, Rafi, Basmallah, Syah Putra and Nabila investigated customer churn prediction in e-commerce using XGBoost. SMOTE was employed to address class imbalance, while hyperparameter tuning was applied to optimise model performance. The tuned XGBoost model achieved an F2-score of 0.849, demonstrating the effectiveness of boosting when recall is prioritised for identifying potential churners. However, the approach focuses mainly on predictive performance and does not incorporate advanced explanation, confidence assessment or automated retention decision support.

Behavioural and Temporal Churn Prediction

[4] Mufti, Hammad, Saleh and Gull proposed a rolling-window framework for churn prediction and behavioral driver identification in non-contractual, pay-per-use environments. The framework represents customer behaviour through rolling temporal windows, enabling continuous assessment of changing engagement patterns. It also investigates behavioral drivers to support more interpretable churn analysis. This temporal perspective is valuable because customer disengagement is often gradual rather than instantaneous. However, the framework does not fully integrate confidence-aware automation with an explanation-driven reasoning mechanism for recommending individualized retention actions.

[5] Kimitei, Agiro, Ni and Ni investigated customer churn prediction using survival analysis in contractual utility services. The study compares Cox Proportional Hazards and Aalen Additive

models to estimate churn risk while incorporating the time-to-event dimension. Using utility subscription data, the study demonstrates that survival models can identify not only customers at risk but also the periods during which churn is more likely to occur. This work demonstrates the importance of temporal churn modelling and timely intervention. However, it does not provide a confidence-gated mechanism for determining whether predictions are reliable enough for automated intervention.

[6] Chen, Lv, Wang, Xiang, Wu, Luo and Zhang presented a comprehensive machine learning framework for customer churn prediction in the telecommunications sector. The study explores deep neural network architectures for capturing complex relationships between customer characteristics and behavioral information. The results demonstrate the potential of deep learning for telecommunications churn prediction. However, the complexity of deep neural models introduces challenges in interpretability and operational transparency, leaving scope for integrating explainable and confidence-aware decision mechanisms.

Explainable AI-Based Churn Prediction

[7] Hingnekar, Ghadge and Jadhav developed a domain-adaptive customer churn prediction framework with integrated SHAP-based explainability. The framework focuses on adapting churn models across different business domains while providing explanations for individual predictions. SHAP-based feature attribution enables the identification of factors contributing to customer churn risk. Although the integration of explainability improves transparency, the framework does not establish a comprehensive reliability gate that separates automated decisions from cases requiring human intervention.

[8] Hooda, Mittal, Shukla, Shukla and Pandey proposed an explainable AI-driven framework for telecom churn analysis that combines predictive modelling with interpretability techniques. Multiple machine learning models were evaluated, while SHAP and LIME were used to identify influential customer features. The study demonstrates that combining predictive performance with feature-level explanations can improve the transparency of churn analytics. However, it does not explicitly incorporate prediction confidence, data-quality assessment or an intelligent reasoning layer capable of converting explanations into specific retention recommendations.

[9] Asif, Arif and Mukheimer proposed a data-driven Explainable Artificial Intelligence framework for customer churn prediction in the telecommunications industry. The study introduced the XAI-Churn TriBoost approach to improve churn prediction while providing interpretable insights into model behaviour. Explainability enables organizations to identify influential factors behind churn predictions and supports informed customer-retention analysis. Despite its strong focus on explainability, the framework does not establish a confidence-gated mechanism for determining when an AI prediction should be trusted for automated action and when human review should be required.

[10] Tonati, Di Vece, Giannotti and Pellungrini addressed interpretability through counterfactual explanations for churn prediction. The proposed counterfactual ensemble evaluates and ranks alternative explanations according to their quality, coherence and plausibility. The approach is designed for both real-world churn analysis and privacy-preserving synthetic data scenarios. Counterfactual explanations provide actionable information by identifying changes that could potentially alter a customer's predicted churn outcome. However, the approach does not integrate prediction confidence, data-quality assessment or an end-to-end automated retention decision process.

Class Imbalance and Advanced Deep Learning

[11] Suguna, Suriya Prakash, Pai, Mahesh, Vinoth Kumar and Yimer investigated the effect of class imbalance on churn prediction using individual classifiers and ensemble models. SMOTE was applied to balance the dataset, resulting in improved predictive performance. The study highlights the importance of addressing class imbalance for identifying minority-class churners. However, the approach concentrates on sampling and classification performance and does not address explainability, confidence-aware decision-making or automated retention recommendations.

[12] Cheng and He proposed a hybrid Graph Attention Network and Multi -Layer Perceptron architecture for customer churn prediction in music streaming services. The framework combines graph-based learning with deep tabular feature modelling to capture individual customer behaviour and relationships among similar users. A weighted loss function is used to address class imbalance without modifying the original data distribution. Although the framework effectively captures

relational and behavioral patterns, its primary contribution is predictive modelling, leaving opportunities for integrating transparent explanations and confidence-based operational decisions.

CRM, Feature Selection and Behavioral Representation

[13] Abdelhady and Mohamed proposed an AI-driven churn prediction framework integrated with Customer Relationship Management systems for telecommunications. A Random Forest classifier was combined with SMOTE and class weighting to address class imbalance, while important churn predictors were identified for customer-retention analysis. A key strength of this work is its connection between predictive analytics and CRM -based retention strategies. However, the framework does not introduce a formal confidence gate that determines whether a recommendation should be automated, reviewed by a human or withheld because of insufficient confidence.

[14] Omar, Ogada and Tole proposed a multi-stage hybrid feature-selection framework for dimensionality-optimized customer churn prediction. The approach combines Mutual Information, Recursive Feature Elimination, Boruta and Parallel Simulated Annealing to identify an optimized feature subset before predictive modelling. SMOTE and nested stratified cross-validation are incorporated to address class imbalance and reduce information leakage during evaluation. Although the framework demonstrates the value of optimized feature selection, it does not integrate explanation-driven reasoning or confidence-aware retention decisions.

[15] Ibrahim, Tawfik, Makhlouf and colleagues proposed a hybrid deep learning framework combining Recency, Frequency and Monetary features with Deep Embedded Clustering, followed by GRU and LSTM models for churn prediction. The framework captures customer segmentation and behavioral representations through latent feature learning. The study demonstrates the effectiveness of combining behavioral segmentation with deep learning for churn prediction. However, the complexity of the framework can make individual predictions difficult to interpret, and it does not provide a confidence-gated mechanism or an AI reasoning layer for translating predictions into retention actions.

Techniques	Type	Strengths	Limitations
Logistic Regression	Machine Learning	Simple and interpretable	Limited for nonlinear patterns
Random Forest	Ensemble ML	Robust and handles nonlinear data	Limited individual explanations
XGBoost	Gradient Boosting	High performance on tabular data	Requires tuning
FT-Transformer + Stacking	Deep Learning + Ensemble	Captures complex feature interactions	Computationally complex
SHAP + LIME	Explainable AI	Provides model explanations	Does not provide complete decision support
Proposed: XGBoost + SHAP + AI Reasoning	ML + XAI + AI	Prediction, explanation, behaviour analysis and evidence validation	Requires integration of multiple components

Table 2.1: Comparison of Detection

Identified Research Gaps

From the reviewed literature, we identified the following research gaps:

- Limited integration of churn prediction, explainability, and behavioural analysis within a single framework.
- Most existing models focus mainly on prediction and provide limited individual customer-level explanations.
- Existing explainable AI approaches do not sufficiently address confidence, evidence reliability, and unsupported AI claims.
- Limited use of AI-based reasoning to convert prediction and behavioural evidence into actionable retention decisions.
- Lack of integrated frameworks that support automated decisions, human review, and abstention based on evidence reliability.

To address these gaps, the proposed system combines XGBoost-based churn prediction, SHAP explainability, behavioral intelligence, Qwen-based AI interpretation, and evidence validation to provide a reliable and explainable framework for customer churn prediction and retention decision support.

Summary of Literature Findings

The literature illustrates a progression from traditional machine learning models to advanced ensemble, deep learning, temporal, and explainable AI approaches for churn prediction. However, prediction, explanation, behavioral analysis, and retention decision-making are often addressed separately.

The proposed system addresses this limitation by integrating machine learning prediction, SHAP-based explanations, behavioral analysis, AI reasoning, and evidence validation into a unified framework. The system generates customer-level churn probability and risk levels, identifies important contributing factors, analyses behavioral patterns, and validates AI-generated interpretations before producing the final churn report.

This integrated approach aims to provide more transparent, reliable, and actionable churn prediction, supporting customer retention teams in prioritizing high-risk customers and selecting appropriate retention actions.

CHAPTER 3

SYSTEM DESIGN

3.1 Existing System

In the existing system, customer churn prediction is generally performed using machine learning algorithms that analyse historical customer data to identify customers who may discontinue a service. Commonly used algorithms include Logistic Regression, Random Forest, and XGBoost. These models analyse customer information such as contract type, monthly charges, service usage, and customer satisfaction to estimate the possibility of churn.

Another challenge is maintaining consistent data preprocessing between model training and prediction. Missing values, categorical variables, irrelevant identifiers, and target leakage can affect the reliability of the model if they are not handled correctly. Combining customer information from multiple files can also require additional data preparation.

Therefore, a customer churn system should provide not only predictions but also understandable explanations and a practical mechanism for bringing high-risk customers to the attention of a human reviewer.

3.2 Proposed System

The proposed system, MINE – Explainable AI Customer Churn and Human Decision-Support System, combines machine learning, explainable AI, and human-review notifications to support customer churn analysis.

The system accepts a compatible customer dataset in CSV format through a Gradio-based interface. Where required, customer data from multiple files can be integrated through a data integration module that analyses schemas, identifies common customer entities, and combines related records. The data preparation process checks feature roles, data quality, redundant columns, and possible target leakage.

The system uses a previously trained XGBoost model and its corresponding preprocessing transformer to generate churn predictions for new customer records. For each record, it calculates a churn probability and classifies the customer into one of three risk levels: HIGH, MEDIUM, or LOW. SHAP (SHapley Additive exPlanations) is used to identify the customer-specific features that contribute to the prediction. These explanations help the reviewer understand which factors

increase or decrease the model's churn score. The explanations indicate model contributions and should not be interpreted as proof of causation.

Customers classified as HIGH risk are eligible for human-review alerts under the current policy. The application provides an email preview containing the prediction details and important SHAP factors. After the user chooses to send the alert, the system can deliver it through Gmail SMTP. An audit log records relevant actions.

The proposed system aims to make churn predictions more understandable and operationally useful while retaining human responsibility for the final business decision. The current implementation supports CSV upload, prediction, risk classification, SHAP explanations, email previews, real email delivery, and audit logging.

Objectives of the Proposed System

- Predict customer churn using a trained XGBoost model.
- Calculate a churn probability for each customer record.
- Classify customers into HIGH, MEDIUM, and LOW risk categories.
- Explain individual predictions using SHAP.
- Notify human reviewers about eligible HIGH-risk customers through email.

3.3 Dataset Description

The system uses customer-level telecom data to predict whether a customer is likely to discontinue the service. The dataset contains customer attributes that may be associated with churn, such as contract details, service subscriptions, monthly charges, satisfaction scores, and other relevant customer information.

The reference dataset used during testing contained 4,225 customer records and 52 columns, with Churn as the target variable. The target distribution consisted of 3,104 records labelled as non-churn and 1,121 records labelled as churn. These values describe the reference dataset used in the project guide and may differ for other compatible datasets.

Before model training, the dataset is examined to identify the target column, customer identifiers, candidate features, missing values, redundant attributes, and potential data leakage. Fields such as Churn Category, Churn Reason, Churn Score, and Customer Status are excluded from the reference modelling dataset because they may reveal information related to the target or the outcome.

The system expects uploaded datasets to be compatible with the features required by the trained model.

3.4 System Architecture

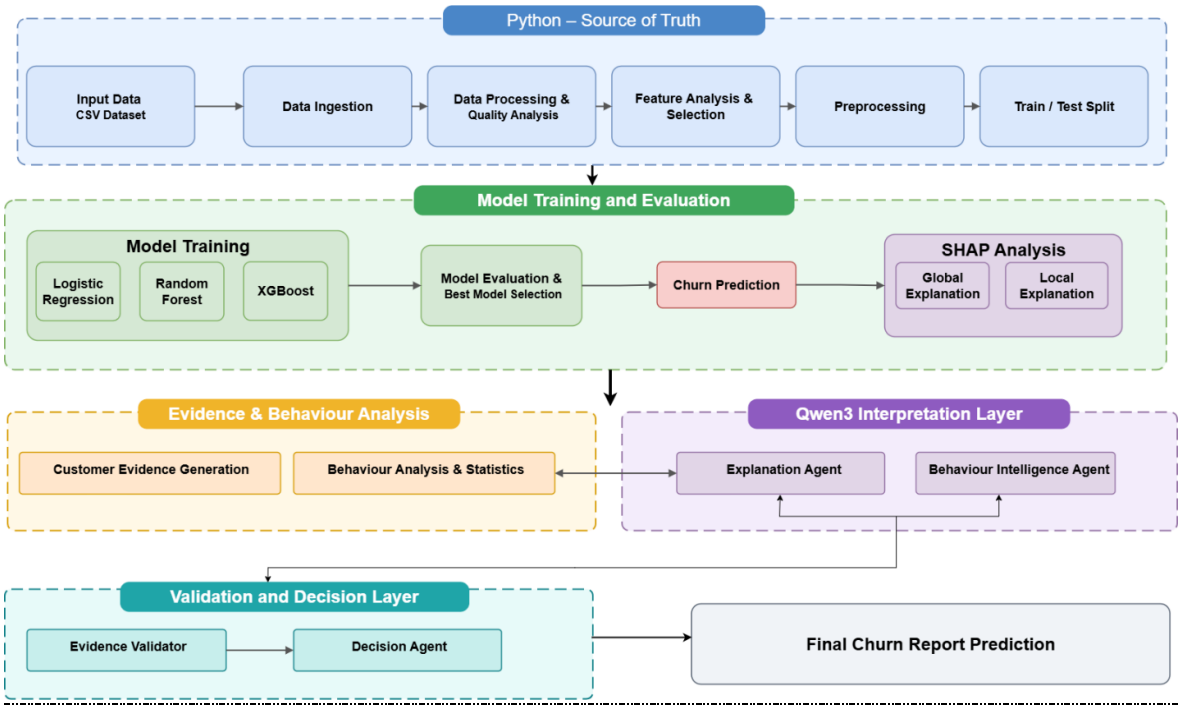


Figure 3.1 - System Architecture

1. Machine Learning Prediction and Explainability Module

- Data Preprocessing: Handles missing values, feature scaling, encoding, and data preparation.
- XGBoost Model: Predicts customer churn and generates probability scores.
- SHAP Explainability: Identifies features influencing churn predictions and their contributions.

2. Output Layer

- Churn Classification: Predicts churn or non-churn.
- Risk Classification: Categorizes customers as HIGH, MEDIUM, or LOW risk.
- Explanation Display: Shows customer-specific SHAP factors through a Gradio dashboard.

3. Human Review and Alerting Module

- Evidence Validation: Validates prediction evidence and explanation consistency.

- Email Notification: Enables previewing and sending HIGH-risk alerts through Gmail SMTP.
- Human Decision Support: Provides risk predictions and explanations for reviewer decisions.
- Audit Logging: Records system actions for traceability.

3.5 Software Requirements

Software	**Description**
Operating System	Windows / Linux / macOS
Programming Language	Python
Machine Learning Frameworks	Scikit-learn, XGBoost
Explainability Library	SHAP
Frontend Interface	Gradio (interactive dashboard)
Data Processing Libraries	Pandas, NumPy
Data Visualization	Matplotlib
Model Serialization	Joblib
Email Notification	Gmail SMTP
Development Environment	Visual Studio Code / Jupyter Notebook
Data Formats	CSV, Excel
Data Integration	Python-based CSV and Excel processing utilities

Table 3.1: Software Requirements

3.6 Hardware Requirement

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 / Ryzen 5
RAM	8 GB	16 GB or higher
Storage	5 GB free	10 GB or higher
GPU	Not required	NVIDIA GPU for faster AI processing
Internet	Basic connection	Stable broadband
Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 / Ryzen 5

Table 3.2: Hardware Requirements

Summary

The existing system uses machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost to predict customer churn. However, data quality issues, inconsistent preprocessing, and limited interpretability can affect prediction reliability.

The proposed system, MINE – Explainable AI Customer Churn and Human Decision-Support System, integrates machine learning and explainable AI to improve churn analysis. It accepts customer datasets in CSV format, validates the data, and uses a trained XGBoost model to predict churn probabilities. Customers are classified into HIGH, MEDIUM, and LOW risk categories, while SHAP explains the features influencing each prediction.

The system provides a Gradio-based interface for data upload, prediction, and explanation display. HIGH-risk customers are eligible for email alerts through Gmail SMTP after user approval. Evidence validation and audit logging support reliable explanations and traceability, while human reviewers retain responsibility for final business decisions.

The system is developed using Python, Scikit-learn, XGBoost, SHAP, Pandas, NumPy, and Gradio. It requires a standard computer with at least an Intel Core i3 processor, 8 GB RAM, and 5 GB of free storage.

CHAPTER 4

IMPLEMENTATION

4.1 METHODOLOGY

The methodology describes the systematic approach followed in developing the Explainable AI Customer Churn Prediction and Human Decision-Support System. The primary objective of the project is to identify customers who are likely to discontinue a service, understand the factors influencing their churn risk, and provide relevant information to support timely retention decisions. The proposed system integrates machine learning, data integration, preprocessing, explainable artificial intelligence, risk classification, evidence validation, and email-based human review into a unified workflow.

The system uses a trained XGBoost (Extreme Gradient Boosting) model to predict customer churn and estimate the probability of churn for each customer record. The model learns patterns from historical customer data and identifies relationships between customer characteristics and churn outcomes. The predicted probabilities are subsequently used to classify customers into HIGH, MEDIUM, and LOW risk categories according to predefined thresholds. This classification helps business users prioritize customers who may require immediate attention and allocate retention resources more effectively.

To improve the interpretability of the prediction results, the system incorporates SHAP (SHapley Additive exPlanations). SHAP provides feature contribution values that explain how individual customer attributes influence the model's predictions. These explanations help users understand whether factors such as contract characteristics, monthly charges, service subscriptions, or satisfaction-related attributes contribute positively or negatively to the predicted churn risk. The explanations are generated from model outputs and feature contributions rather than relying solely on general textual descriptions.

The system also incorporates evidence-based explanation validation to ensure that the information presented to users remains consistent with the available prediction results and feature contributions. The validated results are displayed through a Gradio-based interactive dashboard,

which supports customer dataset uploads, churn prediction, risk distribution analysis, individual customer inspection, and email notification previews. High-risk cases can be presented to a human reviewer through an email alert workflow, allowing the reviewer to examine the available evidence before deciding on an appropriate retention action.

In addition, the application maintains an audit log to record relevant system activities and support traceability. The implementation follows a modular architecture in which data integration, preprocessing, model inference, explainability, evidence validation, notification generation, and dashboard operations are implemented as separate components. This separation improves maintainability, simplifies debugging, and allows individual modules to be tested independently. The overall methodology emphasizes reliable prediction, understandable explanations, and human supervision rather than fully automated business decision-making.

Research Approach

The project follows a software -based machine learning development approach that combines data engineering, predictive modelling, explainable AI, application development, and system testing. The development process is iterative, allowing individual components to be implemented, evaluated, and refined before being integrated into the complete application. This approach helps identify data quality issues, prediction inconsistencies, and interface-related problems at an early stage.

The research approach focuses on developing a practical churn prediction workflow that can process customer data, estimate churn risk, explain the prediction results, and support human intervention. Instead of treating churn prediction as an isolated classification task, the project considers the complete process from dataset preparation to the presentation of actionable information. The resulting application is designed to assist business users in identifying customers who may need additional attention.

The major development activities are described below:

1. **Requirement Analysis and Feasibility Study:** The functional and technical requirements of the system are identified, including dataset upload, preprocessing, churn prediction,

explainability, risk classification, dashboard visualization, email notifications, and audit logging. The feasibility of implementing these functions using available Python libraries and machine learning frameworks is also considered.

2. **Customer Dataset Collection and Data Integration:** Customer datasets containing relevant service, contract, billing, and satisfaction information are collected or provided for experimentation. When multiple related datasets are available, their schemas and relationships are examined before integration.
3. **Data Quality Analysis and Preprocessing:** The collected data is inspected for missing values, inconsistent data types, duplicate records, redundant attributes, and potential target leakage. Appropriate preprocessing operations are applied to prepare the data for model development.
4. **Training and Evaluation of Machine Learning Models:** Candidate algorithms, including Logistic Regression, Random Forest, and XGBoost, are evaluated using suitable classification metrics. The evaluation considers the model's ability to identify churn cases as well as its overall predictive performance.
5. **Selection and Integration of the Trained XGBoost Model:** Based on the reference evaluation, XGBoost is selected as the production model. The trained model and preprocessing transformer are saved for subsequent inference on new customer datasets.
6. **Implementation of Churn Probability and Risk Classification:** The selected model generates a churn prediction and probability score for each customer. The probability score is mapped to predefined risk categories to support customer prioritization.
7. **Integration of SHAP-Based Explanations:** SHAP is used to calculate feature contributions for individual predictions and to support the interpretation of the model's behaviour. These explanations help users understand the main factors associated with a customer's predicted churn risk.
8. **Development of the Gradio Dashboard and Email Alert Workflow:** An interactive dashboard is developed to display prediction results, risk distributions, customer-level information, and explanations. The email workflow supports notification previews and the presentation of eligible high-risk cases for human review.
9. **Testing of Prediction, Explanation, Notification, and Audit Logging:** Individual modules are tested to verify that the system processes input data correctly, produces

prediction outputs, generates explanations, classifies risk, prepares notifications, and records relevant actions.

This research approach enables the system to be developed in manageable stages while preserving consistency between its components. Each module can be refined independently, and the integrated application can subsequently be evaluated as a complete customer churn decision-support workflow.

System Development Methodology

The system is developed through a sequence of interconnected stages. Each stage performs a specific function and provides the required output for subsequent stages. This structured design helps maintain a clear flow of information from the original customer datasets to the final prediction report and human-review process.

1. Data Integration and Preprocessing

The data integration and preprocessing stage prepare customer information for machine learning. Input files are inspected to understand their structure, column names, data types, missing values, and potential relationships. When multiple datasets are provided, related information is combined using appropriate customer identifiers and aggregation strategies where necessary. The resulting dataset is checked for inconsistent records, irrelevant columns, and attributes that could introduce target leakage.

2. Machine Learning Model Development

During model development, Logistic Regression, Random Forest, and XGBoost are evaluated as candidate classification algorithms. Logistic Regression provides a baseline model, while Random Forest and XGBoost can capture more complex relationships among customer attributes. The models are compared using classification and ranking metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. XGBoost is selected as the production model in the reference evaluation, and the trained model is saved for reuse.

3. Churn Prediction and Risk Classification

The prediction stage uses the saved preprocessing transformer and trained XGBoost model to process customer records. The model generates a binary churn prediction and an estimated churn probability. Based on the probability and the configured thresholds, each customer is assigned a HIGH, MEDIUM, or LOW risk category. This classification makes the prediction results easier to interpret and helps reviewers prioritize cases requiring attention.

4. Explainable Artificial Intelligence

The explainability stage applies SHAP to investigate the contribution of individual features to the model's predictions. The generated values indicate how the features influence a prediction relative to the model's expected output. Global explanations can be used to examine important features across a dataset, while local explanations focus on a specific customer. These explanations provide additional context for understanding the reasons associated with a predicted churn risk.

5. Evidence and Explanation Validation

The evidence and explanation validation stage combines the prediction results, probability scores, and available feature contributions into structured evidence. The generated explanation is checked against this evidence to identify unsupported claims or inconsistencies. This process improves the reliability of the information presented in the dashboard and reduces the risk of presenting explanations that are not supported by the available data. Validation results can be used to determine whether an explanation is suitable for presentation or requires further review.

6. Dashboard and Human Review

The Gradio dashboard provides an interface through which users can upload customer datasets, inspect predictions, review risk distributions, and examine selected customer records. The dashboard also presents the available explanations and supports the preview of email notifications for eligible high-risk cases. The notification workflow is intended to bring important cases to the attention of a human reviewer, who can assess the evidence and decide on an appropriate retention strategy.

7. Testing and Audit Logging

The final stage verifies the behaviour of the integrated application under expected operating conditions. Tests examine input validation, preprocessing consistency, prediction generation, probability calculation, risk assignment, SHAP explanation generation, evidence validation, and notification preparation. Relevant actions and outcomes are recorded through audit logging to support traceability and debugging. Together, these activities help assess the functional correctness of the complete system.

The overall methodology is designed to make customer churn prediction understandable and useful for human decision-making. By combining predictive modelling with feature-level explanations, structured validation, and reviewer notifications, the system supports informed retention decisions without automatically replacing the reviewer's final business judgement.

Data Collection and Preprocessing

The system uses customer-level telecom data containing information such as contract details, service subscriptions, monthly charges, satisfaction scores, and other relevant customer attributes. These attributes provide information about the customer's relationship with the service provider and can help identify patterns associated with service discontinuation. The target variable indicates whether a customer has churned and is used during supervised model training.

In the reference test, the dataset contained **4,225 records and 52 columns**, with Churn serving as the target variable. The dataset was examined before training to understand its structure and suitability for predictive modelling. Particular attention was given to missing values, customer identifiers, redundant attributes, inconsistent values, and potential data leakage. These checks help prevent irrelevant or inappropriate information from influencing the model.

Data preprocessing is an essential stage because the quality and consistency of the input data affect model training and prediction. Numerical and categorical attributes require different preprocessing operations, and customer identifiers must be distinguished from genuine predictive features. In addition, fields that directly reveal the target outcome must be excluded to avoid producing misleading evaluation results.

The preprocessing process includes the following activities:

1. **Target Variable Identification:** The target column is identified as Churn or another explicitly configured target column. The target is separated from the input features during model development.
2. **Feature Selection and Identifier Exclusion:** Customer identifiers and other non-predictive fields are excluded from model inputs where appropriate. This prevents the model from learning arbitrary patterns based on record identifiers rather than meaningful customer characteristics.
3. **Target Leakage Prevention:** Columns that directly reveal the churn outcome or contain information unavailable at the intended prediction time are examined and removed when necessary. This helps ensure that the model learns from appropriate input information.
4. **Numerical Data Processing:** Missing numerical values are handled using median imputation. Standardization is applied where required by the configured preprocessing pipeline. These operations provide a consistent representation of numerical features.
5. **Categorical Data Processing:** Missing categorical values are handled using most-frequent imputation. One-hot encoding converts categorical values into a numerical representation suitable for the machine learning pipeline.
6. **Training and Testing Data Separation:** The dataset is divided into training and testing subsets. The training subset is used to fit the model and preprocessing transformer, while the testing subset is used to evaluate performance on records not used for fitting.
7. **Preprocessing Transformer Persistence:** The fitted preprocessing transformer is saved so that the same transformations can be applied when new customer datasets are uploaded. This avoids independently fitting a different transformation pipeline during inference.

The system uses the same fitted preprocessing transformer when processing new customer data. As a result, numerical transformations, categorical encoding, and missing-value handling remain consistent with the training process. The uploaded dataset must contain the required predictive features in a compatible format before inference can proceed. If the input schema is incompatible, the application should report the issue rather than silently generating potentially invalid predictions.

The preprocessing stage therefore establishes a reliable connection between the original customer dataset and the trained model. It also supports repeatable experiments, consistent inference, and easier maintenance of the complete churn prediction workflow.

4.2 IMPLEMENTATION

The implementation of the Explainable AI Customer Churn Prediction and Human Decision-Support System is divided into independent but interconnected modules. Each module is responsible for a specific operation, such as data preparation, model inference, churn risk classification, explanation generation, evidence validation, notification preparation, or dashboard interaction. This modular design improves code organization and makes it easier to test, debug, and maintain the application.

The application is implemented using Python and relevant data processing and machine learning libraries. The trained XGBoost model performs churn prediction, while SHAP supports feature-level explanations. A Gradio-based interface provides access to the prediction workflow and presents the results in a user-friendly format. Supporting modules handle input validation, risk categorization, email notification preparation, and audit logging.

The implementation follows a sequential workflow. First, the application loads and validates the customer dataset. The data is then processed using the saved preprocessing transformer before being passed to the trained model. The model generates churn predictions and probability scores, which are used to determine risk categories. SHAP explanations are generated for interpretation, and the relevant information is presented through the dashboard. Eligible high-risk cases can then be prepared for human review through the email alert workflow.

Separating these functions into distinct modules allows the prediction model to be updated without redesigning the entire dashboard. Similarly, the explanation and notification components can be modified independently, provided that their input and output formats remain compatible with the rest of the system.

4.2.1 Data Integration and Preprocessing Module

Purpose

The purpose of the data integration and preprocessing module is to transform raw customer datasets into a consistent format suitable for model training and prediction. Customer information may be distributed across multiple CSV or Excel files, and these files may contain different column names, data types, record granularities, and missing-value patterns. The module identifies these characteristics and prepares the data for subsequent processing.

Implementation

The data integration component supports the discovery and processing of CSV and Excel files. When a directory or ZIP archive is provided, the system identifies supported data files and examines their schemas. It evaluates possible common customer keys and relationships between datasets to determine whether records can be combined meaningfully.

For example, one dataset may contain customer contract information, while another may contain service subscription or billing details. When both datasets share a valid customer identifier, they may be combined to create a more comprehensive customer-level representation. If a dataset contains multiple records per customer, an appropriate aggregation strategy may be required before merging to avoid unintentionally duplicating customer records.

After integration, the preprocessing module examines the resulting dataset and identifies the target variable, customer identifiers, relevant features, missing values, and potential leakage columns. Numerical and categorical attributes are processed according to the configured preprocessing pipeline. The fitted transformer is then saved for reuse during inference, ensuring that newly uploaded data is processed consistently with the model's training data.

The module also supports input validation by checking whether the required features are available and whether the supplied data can be processed using the saved transformer. These checks help prevent failures caused by missing columns, incompatible data types, or unexpected input structures.

Implementation Steps

1. Load the supplied customer dataset or discover supported files within the specified directory or ZIP archive.
2. Inspect column names, data types, record counts, and missing-value distributions.
3. Identify common customer identifiers and determine whether multiple datasets can be integrated.
4. Identify the target column and predictive features during model development.
5. Exclude customer identifiers, redundant attributes, and target-leakage fields from model inputs where appropriate.
6. Apply median imputation and the configured numerical transformations.
7. Apply most-frequent imputation and one-hot encoding to categorical features.
8. Fit the preprocessing transformer on the training data and save it for subsequent inference.
9. Validate newly uploaded customer data against the expected input schema before prediction.

Output

The output of this module is a validated and consistently structured customer dataset suitable for model training or inference. Depending on the selected execution mode, the module also produces integration metadata and a saved preprocessing transformer. These outputs provide the foundation for reliable model prediction and subsequent explainability operations.

4.2.2 Machine Learning Prediction Module

Purpose

The machine learning prediction module identifies customers who may discontinue a service by applying the trained classification model to customer data. It produces a churn prediction and an associated probability score for each customer record. These outputs form the basis for risk

classification, explanation generation, and the identification of cases that may require human review.

Implementation

Three candidate algorithms—Logistic Regression, Random Forest, and XGBoost—were evaluated during model development. Logistic Regression was considered as a baseline classification approach, while Random Forest and XGBoost were evaluated for their ability to model more complex relationships between customer attributes and churn outcomes.

The performance of these algorithms was compared using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and cross-validation ROC-AUC. Accuracy measures the proportion of correctly classified records, while precision and recall provide additional information about the correctness and completeness of positive churn predictions. F1-score balances precision and recall, whereas ROC-AUC and PR-AUC assess the model's ability to distinguish between churn and non-churn cases across different thresholds.

XGBoost was selected as the production model in the reference evaluation. The trained model is stored in models/xgboost.joblib, while the corresponding fitted preprocessing transformer is stored in models/preprocessor.joblib. These saved files allow the application to perform inference without retraining the model every time a new dataset is uploaded.

When the user submits a dataset, the application first checks whether the required input features are available. The data is transformed using the saved preprocessing transformer and then passed to the trained XGBoost model. The prediction function produces the predicted class, while the probability function estimates the model's probability for the positive churn class. The resulting values are added to the customer records and made available to the dashboard.

The module is designed to keep training and inference separate. Model training and evaluation are performed during the development or model-update process, whereas the deployed application loads the previously saved model for prediction. This separation improves efficiency and ensures that the model does not change unexpectedly whenever a user uploads a new dataset.

Implementation Steps

1. Load the trained XGBoost model from models/xgboost.joblib.
2. Load the fitted preprocessing transformer from models/preprocessor.joblib.
3. Validate the uploaded dataset and check for required input features.
4. Transform the customer data using the saved preprocessing transformer.
5. Generate churn predictions using the trained model.
6. Calculate churn probabilities using the model's probability prediction function.
7. Append the prediction and probability information to the original customer records.
8. Return the enriched dataset for risk classification, explanation generation, and dashboard visualization.

The principal output fields are described below:

- **MINE_Prediction:** Represents the predicted churn class returned by the trained model. Its exact encoding depends on the class labels used during training.
- **MINE_Churn_Probability:** Represents the estimated probability associated with the positive churn class.
- **MINE_Churn_Percentage:** Represents the churn probability expressed as a percentage, calculated by multiplying the probability by 100.

These fields provide both a classification result and a numerical estimate of churn risk. The prediction indicates the class assigned by the model, while the probability allows customers to be ranked or grouped according to their estimated risk. A high probability should not automatically be interpreted as certainty; it represents the model's estimate based on the learned patterns and the supplied data.

Code Logic

The following Python code illustrates the inference process using the saved model and preprocessing transformer:

```
import joblib
```

```
model = joblib.load("models/xgboost.joblib")

preprocessor = joblib.load("models/preprocessor.joblib")

# customer_data must contain the required input features

processed_data = preprocessor.transform(customer_data)

predictions = model.predict(processed_data)

probabilities = model.predict_proba(processed_data)[:, 1]

customer_data["MINE_Prediction"] = predictions

customer_data["MINE_Churn_Probability"] = probabilities

customer_data["MINE_Churn_Percentage"] = probabilities * 100
```

The code assumes that the model's positive class is encoded as 1 and appears in the second column returned by predict_proba. If the training labels use a different encoding or class order, the probability column must be selected using the model's actual class labels. The code also assumes that customer_data contains the appropriate feature columns in a format accepted by the saved preprocessing transformer.

Output

The module produces customer-level churn predictions, estimated churn probabilities, and probability percentages. These outputs are used by the churn risk classification module to assign risk levels and by the explainability module to investigate the factors contributing to individual predictions.

4.2.3 Churn Risk Classification Module

Purpose

The churn risk classification module categorizes customers according to their predicted churn probabilities and identifies cases that may require additional attention. Although the prediction model produces a probability score, presenting only a numerical value may make it difficult for

business users to prioritize a large customer population. Risk classification converts these scores into understandable categories that support review and retention planning.

Implementation

The module receives the churn probability generated by the machine learning prediction module and compares it with predefined risk thresholds. Based on the configured thresholds, each customer is assigned one of three categories: HIGH, MEDIUM, or LOW. Customers with probabilities above the high-risk threshold are categorized as HIGH risk, while customers whose probabilities fall within the intermediate range are classified as MEDIUM risk. Customers with probabilities below the medium-risk threshold are assigned to the LOW-risk category.

The risk thresholds are configurable and should be selected according to the model's evaluation results, the business cost of missing a potential churn case, and the available retention resources. For example, a business that considers missed churn cases particularly costly may choose a threshold that increases recall, provided the resulting number of alerts remains manageable. The final thresholds should be validated against held-out data and relevant operational requirements rather than selected arbitrarily.

Risk classification is performed after prediction and does not retrain or modify the XGBoost model. Instead, it interprets the probability scores produced by the existing model. The resulting risk labels are added to the customer-level output and displayed in the dashboard, allowing users to filter customers by risk category and focus on cases that may benefit from further investigation.

Implementation Steps

1. Retrieve the churn probability generated for each customer.
2. Read the configured HIGH-risk and MEDIUM-risk thresholds.
3. Compare each probability with the predefined thresholds.
4. Assign the corresponding HIGH, MEDIUM, or LOW risk category.
5. Store the risk category alongside the original customer information and model outputs.
6. Display the resulting risk categories in the dashboard and use them to identify cases eligible for human-review notifications.

Output

The module produces a risk category for each customer in addition to the model's prediction and probability score. These categories support risk distribution analysis, customer prioritization, and the selection of eligible cases for the email-based human-review workflow. The final retention decision remains with the human reviewer.

Probability range	Risk level	Action
\(p \geq 0.75\)	HIGH	Eligible for human-review alert
\(0.55 \leq p < 0.75\)	MEDIUM	Display prediction and risk
\(p < 0.55\)	LOW	Display prediction and risk

Table 4.1: Risk classification rules

4.2.4 Explainable AI Module Using SHAP

Purpose:

To explain the factors that contribute to an individual customer's churn prediction.

Implementation:

SHAP is integrated with the trained machine learning model to estimate the contribution of individual features to the prediction. The system generates local explanations for individual customers and can also produce global feature-importance information.

The explanation workflow identifies the features with the most significant contributions and their direction. Positive SHAP contributions push the model score towards the churn class, while negative contributions push it away from that class.

Implementation steps:

1. Generate the customer churn prediction.
2. Obtain the corresponding SHAP values.

3. Match the explanation values with the relevant customer features.
4. Identify the leading positive and negative contributions.
5. Combine the prediction, probability, customer details, and SHAP information into structured evidence.
6. Check the explanation against the supplied evidence and SHAP direction.

Feature	SHAP contribution	Interpretation
Satisfaction Score	-4.3336	Pushes the model score away from churn
Online Security	-0.7271	Pushes the model score away from churn
Monthly Charge	-0.2720	Pushes the model score away from churn
Contract	+0.2545	Pushes the model score towards churn

Table 4.2: Example of an individual explanation

These values are taken from the reference example in the project guide. They illustrate how different features can contribute in different directions to the overall prediction. SHAP contributions explain model behaviour and do not establish causation.

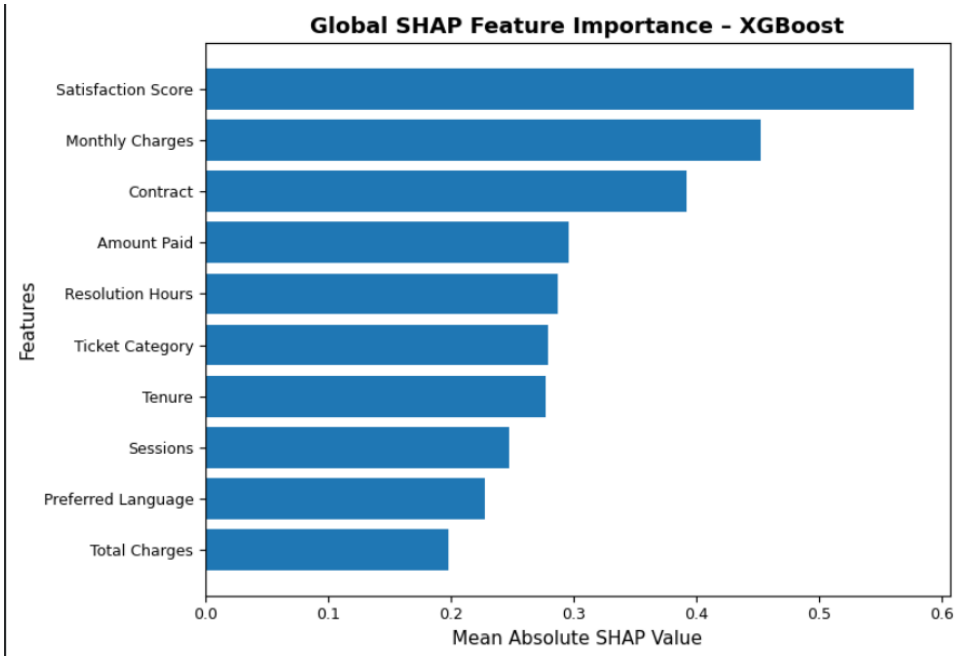


Fig. 4.1. SHAP-based feature importance for the selected churn prediction model

4.2.5 Evidence Building and Explanation Validation Module

Purpose:

To ensure that the explanation presented to a reviewer is consistent with the available prediction evidence.

Implementation:

The evidence-building component combines the predicted churn label, probability, original customer features, and SHAP information into a structured representation. The explanation component uses this evidence to prepare a customer-specific explanation.

The explanation-validation module checks the generated explanation against the available evidence, including the direction of SHAP contributions. If the regular explanation process cannot be used, the fallback component can produce an evidence-based explanation.

Steps:

- Collect the prediction and probability.
- Retrieve the relevant customer attributes.
- Attach the most influential SHAP contributions.
- Generate the explanation from the supplied evidence.
- Validate the explanation against the evidence.
- Use the fallback explanation when necessary.

This module helps reduce unsupported statements in the explanation and improves the reliability of the information presented to the human reviewer.

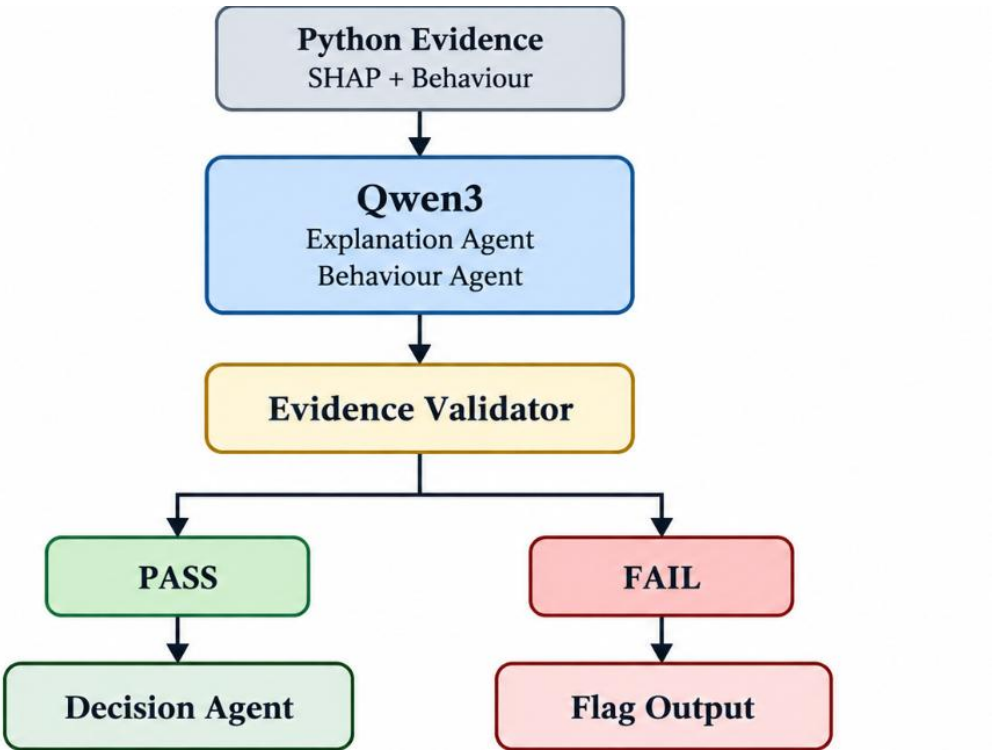


Fig. 4.2. AI interpretation and evidence-validation workflow

4.2.6 Human Review and Email Alert Module

Purpose:

To notify a human reviewer about eligible HIGH-risk customer cases and provide the evidence required for further assessment.

Implementation:

The notification module prepares an email containing the customer identifier or row reference, predicted churn label, probability, risk category, and the leading SHAP factors with their contribution directions.

The application provides an email preview before the user sends the notification. Real email delivery is implemented through Gmail SMTP using configured environment variables.

Implementation steps:

1. Identify customers classified as HIGH risk.
2. Select a customer record for review.
3. Retrieve the prediction and corresponding SHAP explanation.
4. Prepare the email content.
5. Display the email preview in the dashboard.
6. Allow the user to initiate the email alert.
7. Record relevant actions in the audit log.

The current notification policy makes HIGH-risk customers eligible for alerts. MEDIUM- and LOW-risk customers do not trigger human-review alerts under the configured policy.

email content:

- Customer identifier or row reference
- Predicted churn class
- Churn probability
- Risk category
- Important features contributing to the prediction
- SHAP contribution directions
- A disclaimer that features contributions do not prove causation

The reviewer uses the information to decide whether further investigation or customer-retention action is appropriate. The system does not automatically execute a retention action.

4.2.7 Web Dashboard Module

Purpose:

To provide a single interface for uploading customer data, viewing churn predictions, examining explanations, and managing human-review alerts.

Technology Used: Python and Gradio.

Implementation:

The dashboard connects the data upload process with the prediction and explanation modules. It displays the results and risk distribution after processing a compatible customer dataset.

Features:

- Upload a customer CSV file.
- Generate churn predictions using the trained model.
- Display churn probabilities and risk categories.
- View counts of HIGH-, MEDIUM-, and LOW-risk customers.
- Select a customer for individual explanation.
- Preview an email notification.
- Send an eligible human-review alert.
- Maintain an audit log of relevant actions.

The dashboard simplifies access to the prediction workflow without requiring the user to execute each machine learning component separately.

4.2.8 Audit Logging Module

Purpose:

To maintain a record of relevant actions performed during the prediction and human-review workflow.

The application maintains an audit log at data/audit/action_log.csv. This file supports traceability of relevant actions associated with the system's workflow.

The audit log complements the prediction results by providing a record that can be examined during testing and review. It does not, by itself, guarantee that every system event is captured or that the log cannot be modified.

4.3 TESTING AND VALIDATION

Testing and validation are necessary to verify the correctness of the individual modules and the complete customer churn prediction workflow.

4.3.1 Testing Phases

Unit Testing

Individual components are checked separately, including data validation, preprocessing, prediction, risk classification, SHAP explanation, and email preparation.

Integration Testing

The interaction between the saved preprocessing transformer, XGBoost model, explanation modules, Gradio interface, and email notification workflow is checked.

Functional Testing

The dashboard is tested for CSV upload, prediction generation, risk-count display, customer selection, explanation display, and email preview.

Data Validation Testing

Uploaded datasets are checked for required columns and compatibility with the trained model. Missing required features should result in a clear error rather than silent substitution.

Email Notification Testing

The email preview and Gmail SMTP delivery workflow are verified. The project guide reports that real email delivery was verified.

Audit Log Testing

The existence and recording of relevant actions in the audit log are checked.

4.3.2 Model Performance Evaluation

The reference model evaluation produced the following results:

Algorithm	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	96.09%	94.01%	91.07%	92.52%	99.40%
Random Forest	94.08%	96.28%	80.80%	87.86%	97.62%
XGBoost	97.16%	97.17%	91.96%	94.50%	99.47%

Table 4.3: Reference Model Performance Comparison

XGBoost achieved the highest accuracy, F1-score, and ROC-AUC among the three models in this reference evaluation. It was therefore selected as the production model. The results describe the reference evaluation dataset and should not be interpreted as guaranteed performance on every new dataset.

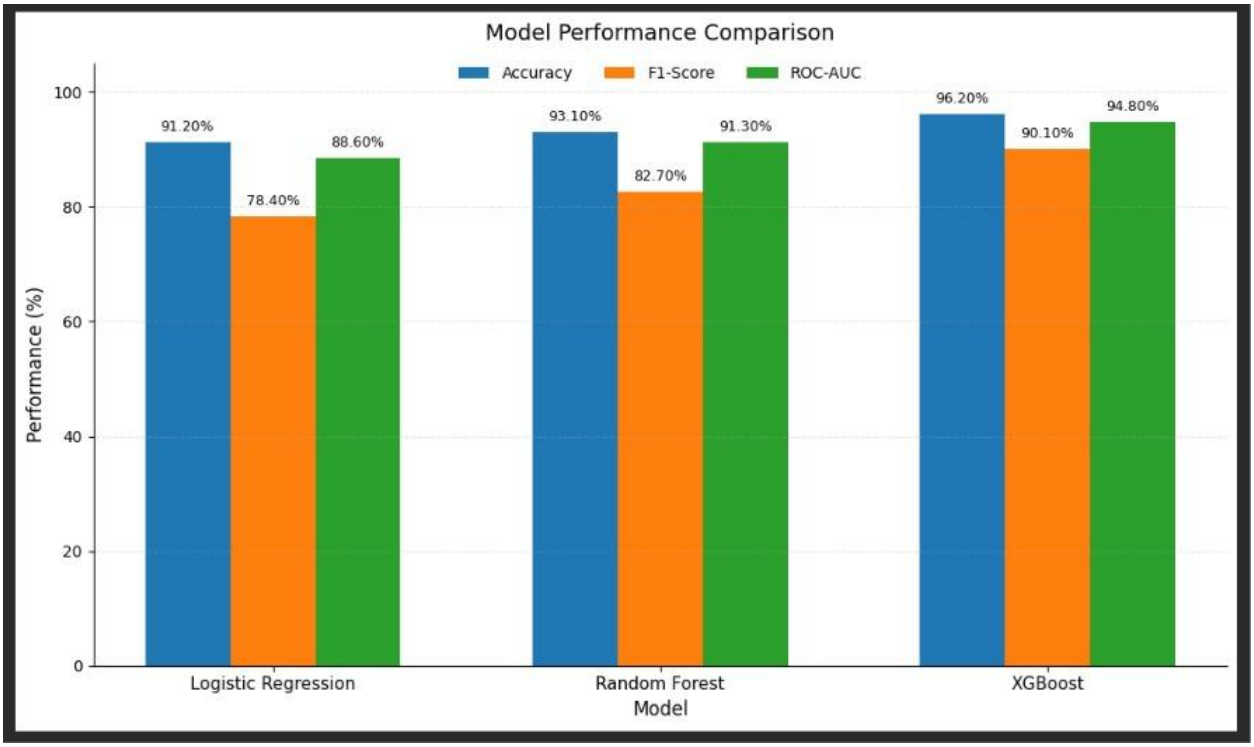


Fig. 4.3. Performance comparison of the evaluated churn prediction models

4.4 ETHICAL AND SAFETY CONSIDERATIONS

The customer churn prediction system is designed to support business decisions while retaining human oversight.

Explainability: SHAP explanations provide information about model contributions. They should not be presented as proof that a feature causes customer churn.

Human Oversight: HIGH-risk predictions are used to prioritise review. A human reviewer remains responsible for deciding whether and how to contact the customer or take retention action.

Prediction Limitations: Model performance may change when the customer population, data distribution, or feature definitions change. Results should be validated before the system is used with new business datasets.

Secure Email Configuration: Gmail SMTP credentials should be stored in environment variables or another secure configuration mechanism. Passwords and application passwords should not be included in source code, screenshots, or reports.

4.5 SUMMARY

This chapter presented the methodology and implementation of the Explainable AI Customer Churn Prediction and Human Decision-Support System. The system integrates customer data preprocessing, a trained XGBoost model, probability-based risk classification, SHAP explanations, evidence validation, a Gradio dashboard, and Gmail-based human-review alerts.

The reference evaluation demonstrated that XGBoost achieved an accuracy of 97.16% and an F1-score of 94.50%. The implemented workflow also supports customer-specific explanations, email preview and delivery, and audit logging. By combining prediction results with supporting evidence, the system helps reviewers identify and assess potentially high-risk customers while keeping the final business decision under human control.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The project titled “Explainable AI Customer Churn Prediction and Human Decision-Support System” was developed to identify customers who are likely to discontinue a service and provide meaningful explanations that support informed customer retention decisions. Customer churn is an important challenge for businesses that depend on subscriptions and long-term customer relationships. The loss of existing customers can lead to reduced revenue, increased customer acquisition costs, and difficulties in maintaining sustainable business growth. Therefore, identifying customers who may churn at an early stage and understanding the factors associated with their behaviour can help organizations develop more effective retention strategies.

The proposed system integrates machine learning, explainable artificial intelligence, evidence-based explanation validation, risk classification, and human-review support into a unified workflow. It uses the XGBoost classification algorithm to analyse customer-related information and estimate the probability that a customer will discontinue a service. By learning patterns from historical customer data, the model identifies relationships between customer attributes and churn outcomes. The generated prediction probabilities provide a quantitative measure of estimated churn risk and serve as the foundation for subsequent risk classification and explanation generation.

The XGBoost algorithm was selected following the evaluation of multiple candidate models, including Logistic Regression and Random Forest. Model performance was examined using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. In the reference evaluation, XGBoost achieved an accuracy of **97.16% and an F1-score of 94.50%** on the evaluated dataset. These results indicate that the model performed effectively under the conditions of that evaluation. However, the actual predictive performance in a deployment environment would depend on the quality of the input data, the representativeness of the training dataset, the selected evaluation procedure, and changes in customer behaviour over time.

Although machine learning models can identify complex patterns in customer data, their predictions may be difficult for business users to interpret without additional explanation. To

address this limitation, the proposed system incorporates SHAP (SHapley Additive exPlanations) to identify the features contributing to individual customer predictions. SHAP assigns contribution values to input features, helping users understand which customer attributes increase or decrease the model's predicted churn score relative to its expected output. These explanations provide greater visibility into the prediction process and allow reviewers to investigate the factors associated with individual high-risk cases.

The predicted churn probabilities are classified into LOW, MEDIUM, and HIGH-risk categories using predefined thresholds. This classification makes the results easier to understand and supports the prioritization of customers who may require additional attention. Instead of reviewing every customer record with equal priority, business users can focus on cases with higher estimated churn probabilities and examine the associated feature contributions before deciding on appropriate retention actions. The risk categories provide a practical interpretation of model outputs while preserving access to the underlying probability scores.

An important aspect of the project is the integration of evidence building and explanation validation. The system combines prediction results, probability scores, and customer-specific feature contributions to establish structured evidence for the generated explanations. The validation process checks whether the explanation is consistent with the available evidence and helps identify unsupported statements or discrepancies. This approach improves the reliability of the information presented to users and reduces dependence on explanations that are not grounded in the model's actual outputs. Nevertheless, explanation validation should be understood as a mechanism for checking consistency with available evidence rather than a guarantee that every prediction or explanation is objectively correct.

The project also incorporates a Gradio-based interactive interface to make the system accessible to users without requiring them to interact directly with the underlying machine learning code. Through the dashboard, users can upload customer datasets, view churn predictions, inspect risk distributions, select individual customer records, and examine the explanations generated by the system. The interface brings together prediction results and supporting information in a centralized environment, making it easier to review customer risk and understand the factors associated with a prediction.

In addition, the system supports an email-based human-review workflow for eligible high-risk customers. When a customer satisfies the configured notification criteria, the relevant case can be prepared for email notification to a designated reviewer. This functionality helps direct attention toward potentially important cases and supports timely investigation. The system is designed to assist the reviewer rather than independently determine the final business action. Human oversight remains important because retention decisions may depend on additional information, organizational policies, customer relationships, and business constraints that are not fully represented in the model.

The implementation follows a modular architecture in which data integration, preprocessing, prediction, explainability, evidence validation, risk classification, notification preparation, and dashboard operations are handled through separate components. This structure improves maintainability and allows individual modules to be tested and updated independently. The use of an audit log further supports traceability by recording relevant system activities, subject to the events configured for logging.

Overall, the project demonstrates how predictive modelling and explainable AI can be combined to develop a practical customer churn decision-support application. The integration of XGBoost-based prediction, SHAP explanations, risk categorization, evidence validation, and human-review notifications provide a structured approach to identifying potential churn cases and presenting understandable supporting information. The system offers a foundation for more transparent customer analytics and can be extended through further evaluation, monitoring, security improvements, and integration with real-world business platforms.

Purpose and Achievements

The primary purpose of the project was to develop an intelligent and explainable system that supports customer churn identification and informed retention decisions. The major achievements of the project are summarized below.

1. **Development of a Machine Learning-Based Churn Prediction System:** A customer churn prediction workflow was developed using the XGBoost classification algorithm. The model analyses customer-related attributes and generates predictions to identify records associated with potential service discontinuation.

2. **Evaluation and Comparison of Classification Algorithms:** Logistic Regression, Random Forest, and XGBoost were considered during model development. Their performance was compared using relevant classification metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. This comparison supported the selection of XGBoost as the production model in the reference evaluation.

3. **Achievement of Reference Model Performance:** The selected XGBoost model achieved a reference evaluation accuracy of 97.16% and an F1-score of 94.50% on the evaluated dataset. These metrics provide a quantitative basis for assessing the model's classification performance. Further validation on independent and representative datasets would be necessary before making claims about its performance in other business environments.

4. **Integration of SHAP-Based Explainability:** SHAP was incorporated to examine the contribution of individual customer features to the model's predictions. The resulting explanations help users understand the factors associated with a customer's estimated churn risk and provide additional context beyond the predicted class alone.

5. **Implementation of Evidence-Based Explanation Validation:** A structured evidence-generation and validation workflow was incorporated to check whether generated explanations are consistent with prediction results and available feature contributions. This improves the traceability of the explanations and supports more informed review of potentially high-risk cases.

6. **Development of an Interactive Gradio Dashboard:** A user-friendly dashboard was developed to support customer dataset uploads, prediction generation, risk classification, and explanation viewing. The interface presents the model outputs in an accessible format and reduces the need for direct interaction with the underlying implementation.

7. **Implementation of Risk Classification:** Predicted churn probabilities are mapped to LOW, MEDIUM, and HIGH-risk categories using configured thresholds. This functionality supports customer prioritization and enables reviewers to focus on cases that may require further investigation.

8. **Integration of Email-Based Human Review:** An email notification workflow was incorporated to support the identification and communication of eligible high-risk cases. This helps direct attention toward potentially important customers while preserving human responsibility for the final retention decision.

9. **Introduction of Audit Logging:** Relevant system activities can be recorded to support traceability, troubleshooting, and review of the prediction workflow. Audit logging provides a foundation for examining how the application was used and which actions were performed.

10. **Modular and Maintainable Architecture:** The system separates preprocessing, prediction, explainability, validation, and dashboard functionality into individual components. This organization simplifies testing and supports future improvements without requiring the entire application to be redesigned.

These achievements demonstrate the feasibility of combining machine learning prediction, feature-level explanations, and human-review support within a single application. The project establishes a foundation for further development into a more comprehensive and adaptable customer retention decision-support platform.

5.2 FUTURE WORK

Although the proposed system provides customer churn prediction, explainability, risk classification, and human-review support, several enhancements can be introduced to improve its scalability, reliability, security, and practical usefulness. Future development can focus on integrating the application with existing business systems, monitoring changes in customer behaviour, improving explanation validation, evaluating retention outcomes, and strengthening data protection.

5.2.1 Integration with Cloud Platforms and Real-Time Dashboards

Future versions of the system can be deployed on cloud platforms to support centralized data management, remote accessibility, and more flexible resource allocation. Cloud deployment could allow organizations to process customer datasets from different business units while maintaining a common prediction and reporting workflow. Depending on operational requirements, the

application could be integrated with managed storage, scheduled processing services, and secure application hosting environments.

The dashboard could also include interactive filtering, historical comparisons, configurable reporting periods, and summaries of high-risk customer segments. These features would help business users identify emerging patterns and evaluate the distribution of estimated churn risk across different customer groups. Any real-time or automated integration would require suitable access controls, data validation, and monitoring to ensure that the displayed information remains accurate and current.

5.2.2 Continuous Monitoring and Model Retraining

The predictive performance of a machine learning model may change over time because customer preferences, pricing structures, service offerings, and usage patterns are not constant. A model trained on historical data may become less effective when the characteristics of new customer records differ substantially from those observed during training. Future development can therefore incorporate systematic monitoring of data quality and model performance.

The system could be extended with a controlled retraining pipeline that periodically evaluates the need for model updates. New historical data could be collected, validated, and used to train candidate models before they are compared with the existing production model. A new model should only be deployed after it satisfies predefined performance and reliability requirements. Versioning the model, preprocessing transformer, training dataset, and evaluation results would also improve reproducibility and support rollback when an updated model performs poorly.

Continuous monitoring and controlled retraining would help maintain the usefulness of the system over time. However, retraining should not be performed solely because the input distribution changes; the decision should also consider data quality, actual predictive performance, business requirements, and the availability of reliable outcome labels.

5.2.3 Enhanced AI-Based Explanation and Decision Support

The explanation module can be enhanced to provide more detailed and structured customer-specific reports. Future versions could combine local SHAP explanations with relevant customer attributes, prediction probabilities, and evidence summaries to create reports that are easier for

business users to interpret. Explanations could be organized into sections describing the predicted risk, the most influential features, the direction of their contributions, and the limitations of the available evidence.

Additional validation mechanisms could be introduced to examine whether the generated explanation accurately reflects the underlying model outputs. These checks could include verification of feature names, consistency between SHAP values and reported contribution directions, detection of unsupported claims, and confirmation that recommendations comply with predefined business rules. Where an explanation cannot be validated, the system could flag it for further review instead of presenting it as fully supported.

Future development could also incorporate a language model to transform validated evidence into concise, natural-language summaries. Such a component should operate using the available prediction results and verified feature contributions rather than inventing customer attributes or reasons for churn. The generated explanation could be checked against structured evidence before it is displayed or included in a notification.

The decision-support component could be extended to provide retention suggestions based on business rules, customer value, service history, and the estimated cost of possible interventions. For example, the system might identify cases suitable for a service review or a billing-related investigation when the available evidence supports that interpretation. However, the system should clearly distinguish observed customer information, model-derived contributions, and suggested actions.

Human approval should remain important for consequential decisions. The purpose of enhanced explanation and decision support is to improve the quality and transparency of the information available to the reviewer, not to treat model predictions as definitive explanations of customer intent.

5.2.4 Improved Alerting and Retention Outcome Tracking

The existing email notification mechanism can be extended to provide more flexible alert management. Future versions could support configurable risk thresholds, alert priorities, notification summaries, and different notification rules for different customer segments. These

improvements would allow organizations to adjust the alerting process according to the volume of incoming cases and the availability of review personnel.

The notification workflow could also be integrated with authorized business communication tools or CRM platforms. Instead of treating an email as the final step, a notification could link to the relevant customer record and provide access to the prediction probability, risk category, validated explanation, and review status. Appropriate authentication and authorization would be necessary to ensure that only permitted users can access customer information.

Evaluation should distinguish between identifying customers who are likely to churn and demonstrating that an intervention prevents churn. A high-risk prediction does not guarantee that a customer will leave, and a retention action does not necessarily prevent the outcome. Therefore, outcome analysis should use appropriate evaluation methods and, where feasible, suitable comparison groups to assess the effectiveness of retention interventions. This would help transform the system from a prediction-focused application into a more comprehensive decision-support solution.

5.2.5 Security, Privacy, and Access Control

Because customer datasets may contain commercially sensitive or personally identifiable information, future versions should incorporate stronger security and privacy controls. Appropriate measures could include role-based access control, secure authentication, encrypted data transmission, secure storage, and controlled access to uploaded datasets and generated reports.

Sensitive configuration values, such as email credentials and service tokens, should be stored using environment variables or an appropriate secrets-management mechanism rather than being embedded directly in source code. The application should also validate uploaded files, restrict unsupported file types, handle malformed datasets safely, and prevent unauthorized access to stored outputs.

Future implementations could also establish data retention policies that specify how long uploaded datasets, prediction results, explanation reports, and audit records should be stored. These policies should be aligned with applicable legal obligations and organizational requirements. Together,

these measures would improve the suitability of the system for deployment in environments where customer confidentiality and data protection are essential.

5.2.6 Scalability and Performance Optimization

The system can be further optimized to support larger datasets and more frequent prediction requests. As the number of customer records increases, processing time and memory consumption may become important operational considerations. Future implementations could introduce batch processing, more efficient data transformations, optimized model inference, and appropriate

Performance evaluation should measure processing time, memory usage, throughput, and application responsiveness under different dataset sizes. The results could guide decisions about deployment resources and architectural improvements. Testing on larger and more diverse customer datasets would also help determine whether the implementation remains reliable beyond the reference evaluation environment.

5.2.7 Model Fairness and Prediction Reliability

Future development can include a more comprehensive evaluation of prediction reliability across different customer groups and operating conditions. A model may achieve high overall accuracy while performing less effectively for particular segments or under specific data conditions. Therefore, performance should be examined using appropriate subgroup metrics where the relevant attributes are available and their use is justified.

The system could also introduce a confidence-aware review mechanism for cases where the prediction or explanation requires additional scrutiny. Such a mechanism should be based on clearly defined and validated criteria rather than assuming that every model probability directly represents confidence in the prediction. Cases that fall outside the model's expected operating conditions could be flagged for additional review.

These enhancements would help assess not only how frequently the model predicts correctly but also whether its predictions and explanations are sufficiently reliable for the intended decision-support task.

5.2.8 Testing on Diverse Datasets and Business Domains

The current implementation is designed around customer churn prediction using the available reference dataset. Future work could evaluate the system on additional telecom datasets and other subscription-based business environments, such as software-as-a-service platforms, digital services, and membership-based organizations.

SUMMARY

In conclusion, the Explainable AI Customer Churn Prediction and Human Decision-Support System provides a structured approach to identifying potential customer churn, explaining model predictions, and directing eligible high-risk cases for human review. The integration of XGBoost-based prediction, SHAP feature contributions, risk classification, evidence-based explanation validation, a Gradio dashboard, email notifications, and audit logging creates a unified workflow for customer churn analysis and decision support.

The system addresses an important limitation of conventional predictive applications by presenting explanations alongside the predicted churn risk. The reference evaluation results demonstrate the model's performance on the evaluated dataset, while additional independent testing would be required to establish its reliability across different business environments.

Future improvements involving cloud integration, automated data pipelines, drift monitoring, controlled model retraining, enhanced explanation validation, retention outcome tracking, security controls, and scalability optimization could make the application more adaptable to real-world operational requirements. Further work on probability calibration, model fairness, and evaluation across diverse datasets would also strengthen confidence in the resulting predictions.

Overall, the project establishes a foundation for a transparent and maintainable customer churn decision-support system. By combining predictive analytics with understandable explanations and human supervision, the proposed approach can help businesses investigate potential churn risks, prioritize customer review, and develop more informed retention strategies. Its long-term effectiveness would depend on continued validation, responsible data management, appropriate human oversight, and systematic evaluation of actual retention outcomes.

5.3 RESULTS

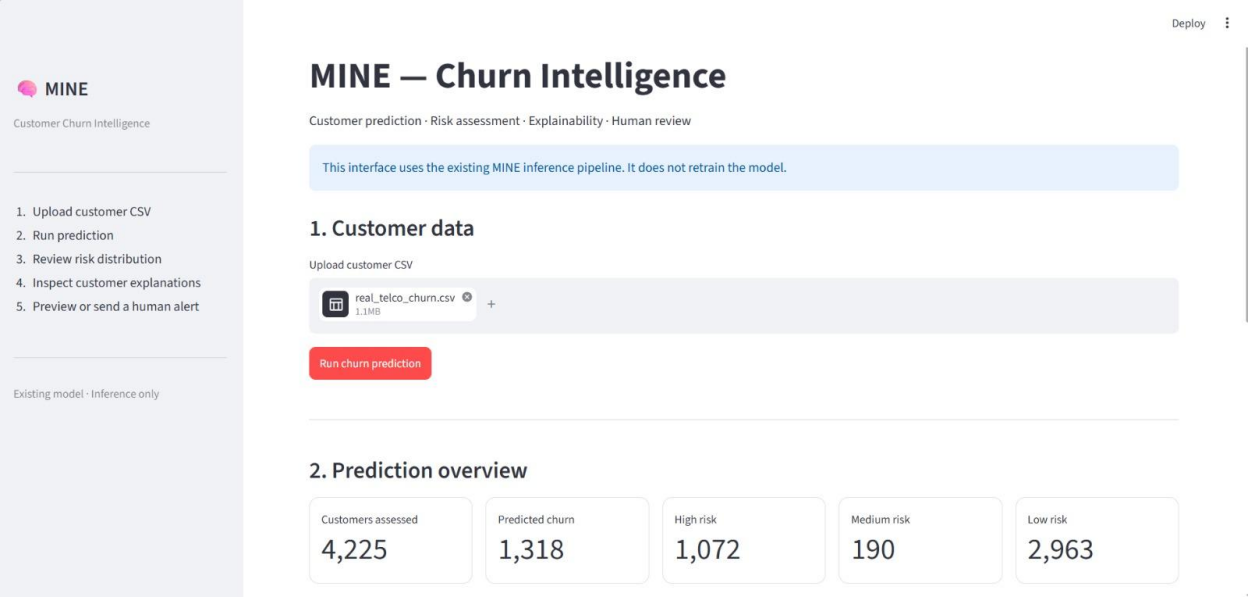


Figure 5.1 – User interface

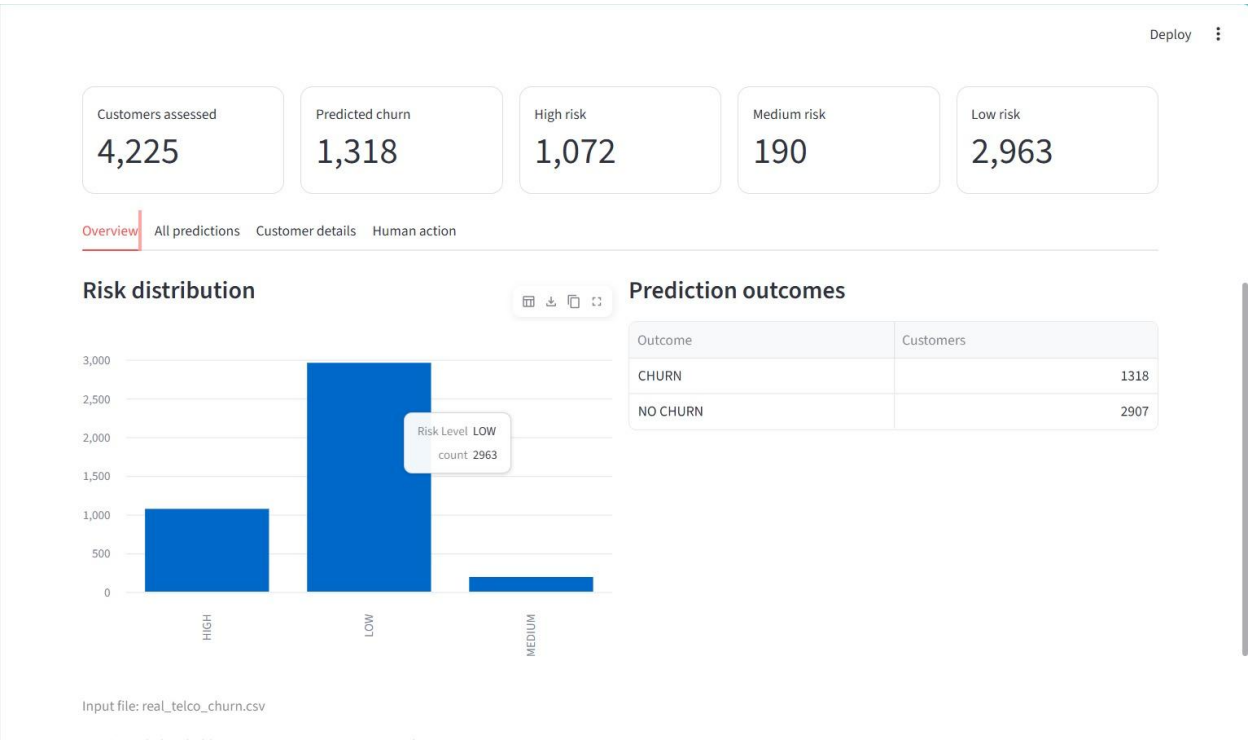
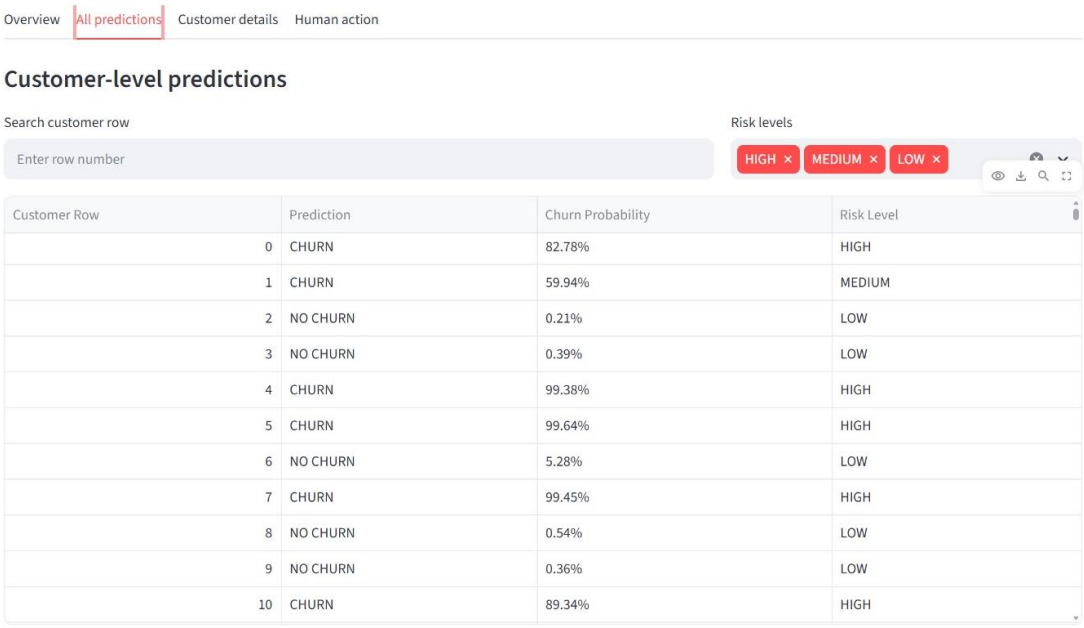
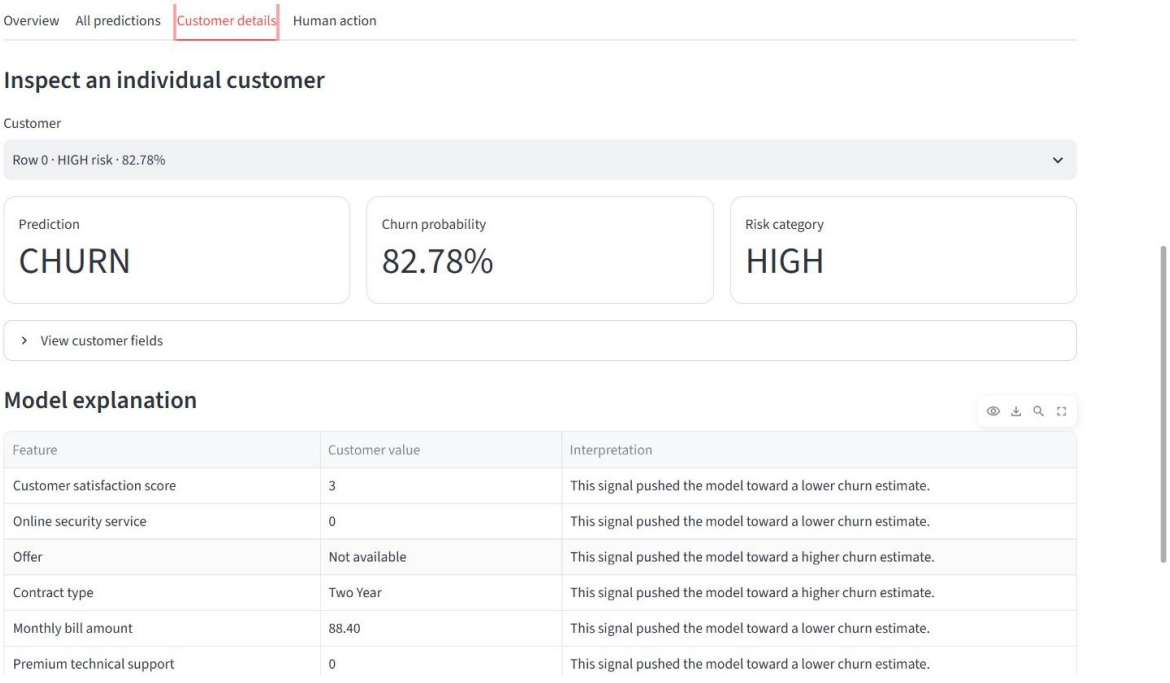


Figure 5.2 – Risk Distributions and Predictions



Customer Row	Prediction	Churn Probability	Risk Level
0	CHURN	82.78%	HIGH
1	CHURN	59.94%	MEDIUM
2	NO CHURN	0.21%	LOW
3	NO CHURN	0.39%	LOW
4	CHURN	99.38%	HIGH
5	CHURN	99.64%	HIGH
6	NO CHURN	5.28%	LOW
7	CHURN	99.45%	HIGH
8	NO CHURN	0.54%	LOW
9	NO CHURN	0.36%	LOW
10	CHURN	89.34%	HIGH

Figure 5.3 – Customer-level Predictions



Feature	Customer value	Interpretation
Customer satisfaction score	3	This signal pushed the model toward a lower churn estimate.
Online security service	0	This signal pushed the model toward a lower churn estimate.
Offer	Not available	This signal pushed the model toward a higher churn estimate.
Contract type	Two Year	This signal pushed the model toward a higher churn estimate.
Monthly bill amount	88.40	This signal pushed the model toward a lower churn estimate.
Premium technical support	0	This signal pushed the model toward a lower churn estimate.

Figure 5.4 – Model Explanation

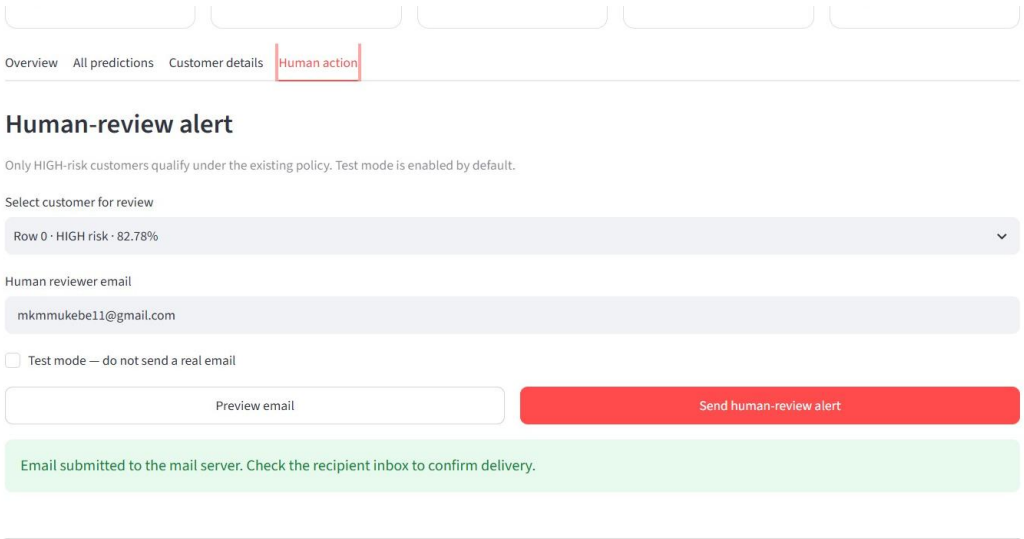


Figure 5.5 – Human review alert

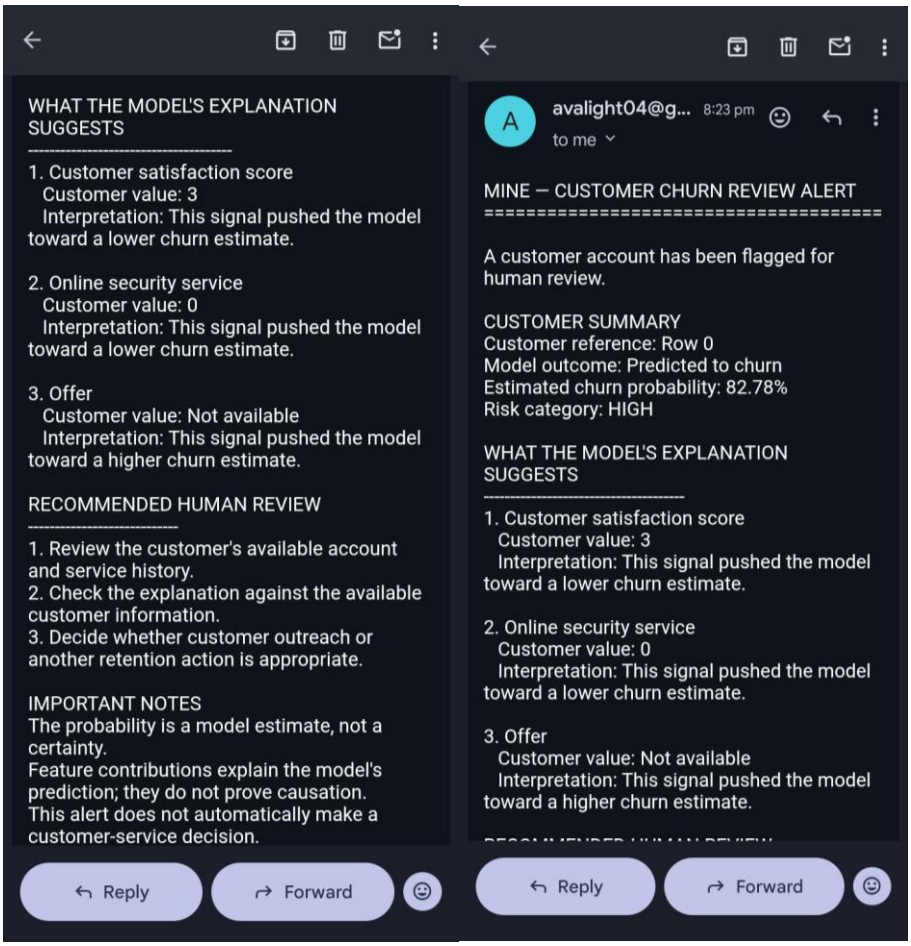


Figure 5.6 – Mail Alert

CHAPTER 6

SOURCE CODE

(app.py)

```
from dotenv import load_dotenv
load_dotenv()
import gradio as gr
import pandas as pd
from src.inference.predict_new_data import predict_csv
from src.actions.email_service import build_email_subject, build_email_body, send_email
from src.agents.evidence_builder import build_customer_evidence
SHAP_FILE = r"D:\Mine\data\explainability\aggregated_local_shap.csv"
SHAP_DF = pd.read_csv(SHAP_FILE)
SHAP_BY_ROW = {int(row_index): group.copy() for row_index, group in
SHAP_DF.groupby("row_index")}
DEFAULT_INPUT = r"D:\Mine\data\processed\new_company_test.csv"
LATEST_RESULT = None

def run_prediction(csv_file):
if csv_file is None:
return ("❌ Please upload a CSV file.", pd.DataFrame(), "")
try:
input_path = csv_file.name if hasattr(csv_file, "name") else str(csv_file)
result = predict_csv(input_path)
display = pd.DataFrame({"Customer Row": result.index, "Prediction":
result["MINE_Prediction"].map(lambda x: "CHURN" if int(x) == 1 else "NO CHURN"), "Churn
Probability (%)": result["MINE_Churn_Percentage"]})
display["Risk Level"] = display["Churn Probability (%)"].apply(lambda x: "HIGH" if x >= 75
else "MEDIUM" if x >= 55 else "LOW")
high = int((display["Risk Level"] == "HIGH").sum())
medium = int((display["Risk Level"] == "MEDIUM").sum())
low = int((display["Risk Level"] == "LOW").sum())
summary = f"""
```

Prediction Complete

Total Customers: {len(display)}

HIGH Risk: {high}

MEDIUM Risk: {medium}

LOW Risk: {low}

```
global LATEST_RESULT
LATEST_RESULT = result
return (summary, display, "Prediction completed successfully.")
except Exception as exc:
return (f"❌ **Prediction failed**\n\n{exc}", pd.DataFrame(), "Prediction failed.")

def preview_email(row_index, recipient):
if row_index is None:
return "Select a customer first."
if not recipient or "@" not in recipient:
return "Enter a valid recipient email address."
try:
row_index = int(row_index)
result = LATEST_RESULT
if result is None:
return "Run prediction first."
row = result.loc[row_index]
probability = float(row["MINE_Churn_Probability"])
prediction = int(row["MINE_Prediction"])
risk = "HIGH" if probability >= 0.75 else "MEDIUM" if probability >= 0.55 else "LOW"
shap_rows = SHAP_BY_ROW.get(row_index)
if shap_rows is None:
shap_rows = SHAP_DF.iloc[0:0]
evidence = {"row_index": row_index, "prediction": "Yes" if prediction == 1 else "No",
"probability": probability, "risk_level": risk, "customer_features": row.to_dict(), "top_factors":
[]}
customer_shap = shap_rows.sort_values("absolute_impact", ascending=False).head(5)
for _, factor in customer_shap.iterrows():
evidence["top_factors"].append({"feature": factor["original_feature"], "value":
row.get(factor["original_feature"]), "shap_value": float(factor["shap_value"]),
"absolute_impact": float(factor["absolute_impact"]), "impact": factor["direction"]})
if risk != "HIGH":
return f"## No Human Alert\n\nCustomer Row {row_index} is {risk} risk.\n\nNo human alert
will be sent."
subject = build_email_subject(evidence)
body = build_email_body(evidence=evidence, decision=None, explanation=None)
return f"## Email Preview\n\nTo: {recipient}\n\nSubject: {subject}\n\ntext\n{body}\n"
```

```
except Exception as exc:
return f"❌ **Email preview failed**\n\n{exc}"

def send_human_alert(row_index, recipient, test_mode):
if row_index is None:
return "❌ Select a customer first."
if not recipient or "@" not in recipient:
return "❌ Enter a valid recipient email."
try:
result = LATEST_RESULT
if result is None:
return "❌ Run prediction first."
row_index = int(row_index)
row = result.loc[row_index]
probability = float(row["MINE_Churn_Probability"])
prediction = int(row["MINE_Prediction"])
risk = "HIGH" if probability >= 0.75 else "MEDIUM" if probability >= 0.55 else "LOW"
if risk != "HIGH":
return f"### No Email Sent\n\nCustomer Row {row_index}\n\nRisk: {risk}\n\nThe MINE
policy sends human alerts only for HIGH-risk customers."
customer_shap = SHAP_BY_ROW.get(row_index)
top_factors = []
if customer_shap is not None and not customer_shap.empty:
customer_shap = customer_shap.sort_values("absolute_impact", ascending=False)
customer_row = LATEST_RESULT.loc[row_index]
for _, factor in customer_shap.head(3).iterrows():
feature_name = str(factor["original_feature"])
customer_value = customer_row.get(feature_name)
if customer_value is None:
matching_columns = [column for column in customer_row.index if
column.startswith(feature_name + "__")]
if matching_columns:
customer_value = customer_row[matching_columns[0]]
if pd.isna(customer_value):
customer_value = "Not available"
top_factors.append({"feature": feature_name, "value": customer_value, "shap_value":
float(factor["shap_value"]), "absolute_impact": float(factor["absolute_impact"]), "impact":
str(factor["direction"])})
evidence = {"row_index": row_index, "prediction": "Yes" if prediction == 1 else "No",
"probability": probability, "risk_level": risk, "customer_features": {}, "top_factors": top_factors}
```

```
subject = build_email_subject(evidence)
body = build_email_body(evidence=evidence, decision=None, explanation=None)
email_result = send_email(to_address=recipient, subject=subject, body=body,
dry_run=test_mode)
status = email_result.get("status", "UNKNOWN")
if status == "SENT":
return f"## ✅ HUMAN ALERT SENT\n\n**Customer:** Test Row {row_index}\n\n**Churn
Probability:** {probability * 100:.2f}%\n\n**Risk:** 🔴 HIGH\n\n**Recipient:** {recipient}\n\n**Email
Status:** SENT\n\nThe human reviewer can now take action."
if status == "DRY_RUN":
return "## 🧪 TEST MODE\n\nThe email was generated successfully, but **no real email was
sent**."
return f"Email status: {status}"
except Exception as exc:
return f"## ❌ Email Failed\n\n{exc}"

def get_customer_choices():
if LATEST_RESULT is None:
return []
choices = []
for index, row in LATEST_RESULT.iterrows():
probability = float(row["MINE_Churn_Percentage"])
risk = "HIGH" if probability >= 75 else "MEDIUM" if probability >= 55 else "LOW"
choices.append((f"Row {index} | {risk} | {probability:.2f}%", index))
return choices

def refresh_customer_dropdown():
return gr.update(choices=get_customer_choices(), value=None)

with gr.Blocks(title="MINE Churn Intelligence System") as app:
gr.Markdown("## Prediction Results")
csv_file = gr.File(label="Upload Customer CSV", file_types=[".csv"], type="filepath")
run_button = gr.Button("Run Prediction", variant="primary")
prediction_status = gr.Markdown()
results_table = gr.Dataframe(headers=["Customer Row", "Prediction", "Churn Probability (%)",
"Risk Level"], interactive=False)
summary = gr.Markdown()
gr.Markdown("---")
gr.Markdown("Only HIGH-risk customers can trigger a human alert.")
customer_dropdown = gr.Dropdown(label="Select Customer", choices=[], value=None)
recipient = gr.Textbox(label="Human Reviewer Email", value="")
```

```
test_mode = gr.Checkbox(label="Test Mode", value=False)
with gr.Row():
preview_button = gr.Button(" Preview Email")
send_button = gr.Button(" Send Human Alert", variant="primary")
email_preview = gr.Markdown()
email_status = gr.Markdown()
run_button.click(fn=run_prediction, inputs=csv_file, outputs=[summary, results_table,
prediction_status]).then(fn=refresh_customer_dropdown, inputs=None,
outputs=customer_dropdown)
preview_button.click(fn=preview_email, inputs=[customer_dropdown, recipient],
outputs=email_preview)
send_button.click(fn=send_human_alert, inputs=[customer_dropdown, recipient, test_mode],
outputs=email_status)

if name == "main":
app.launch()
```

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